



Master Presenter's Manual

for

Embedded Programming

Curriculum Code: OV-1515

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1.0	February 2024	New	-

Table of Contents

Sr. No.	Topic	Page No.
1.	Introduction	1
2.	Information on Session Schedule	2
3.	Week-wise Session Schedule	4
4.	Deliverables	4
5.	Evaluation Strategy	5
6.	Classroom and Lab Pre-requisites	6

1. Introduction

This Presenter's Manual is to be used for Embedded Programming curriculum. The syllabus aims at introducing students to Introduction to Embedded Systems and Programming Embedded System.

The student will learn the following modules in this course:

- Introduction to Embedded Systems
- Programming Embedded Systems

The curriculum contents are specially designed for those who are keen on getting themselves a career in the field of Embedded Systems and programming.

After the completion of this course, students will be able to:

- Learn to implement assembly programming logic, define and use Boolean Operations, perform looping and branching operations, subroutine and interrupt, data movement and arithmetic operations, number system and data operations, and 8051 assembly programming
- Learn the ADC and DAC programming interfaces
- Develop program for I/O ports, Timers and Interrupts

2. Information on Session Schedule

The table below shows the details of the modules, the number of hours being spent on each module along with a break-up of the hours being spent on each module.

This course is based on theory and practice lab session. Each session will be of two hours duration.

Module Wise Session Allocation

Module	Online Session (No. of Hrs)	Skill Code
ITES	40	349
PES	40	349
Total	80	

ITES - Introduction to Embedded Systems
PES - Programming Embedded Systems

➤ OnlineVarsity Deliverables

OnlineVarsity is a digital content access and collaboration platform for Aptech students. The numerous interesting features of the platform will enhance learning experience of the students.

(Faculties can access eBooks and other components in OnlineVarsity through Aprack.)

Resources available on OnlineVarsity (as applicable):

Icons	Feature - Description/Functionality
	Download Book - Student has the option to download the subject related e-book and read offline.
	Glossary - Student can access a list of subject related specialized words with their definitions.
	FAQ - Student can access frequently asked questions and their answers.
	Practice 4 Me - Student can test and evaluate their understanding of module related topics.
	Work Assignments - Student can solve scenario based lab assignments (Hands-on). The faculty will evaluate and give their feedbacks.
	References - Student can access additional subject related material for reading.
	Feedback - Student can provide feedback on the course material.
	Ask to Learn – Student can submit subject related technical queries. Queries submitted will be directed to the particular course coordinator/head.

3. Week-wise Session Schedule

- A Session is of 2 hours duration

➤ WEEK WISE SCHEDULE

Topic\Week		1	2	3	4	5	6	7	8	9	10
Introduction to Embedded Systems	D1	TL1	TL5	TL9	TL13	TL17					
	D2	TL2	TL6	TL10	TL14	TL18					
	D3	TL3	TL7	TL11	TL15	TL19					
	D4	TL4	TL8	TL12	TL16	TL20					
Programming Embedded Systems	D1						TL1	TL5	TL9	TL13	TL17
	D2						TL2	TL6	TL10	TL14	TL18
	D3						TL3	TL7	TL11	TL15	TL19
	D4						TL4	TL8	TL12	TL16	TL20

Where,
TL = Online Session

4. Deliverables

The deliverables to be provided to the students and the faculty have been tabulated below:

➤ Deliverables for Students:

Module	Deliverable
Introduction to Embedded Systems	Kit Code: OV-1515-KIT01-01-B
Programming Embedded Systems	Kit Name: Kit of 2 titles for OV-1515 Embedded Programming

➤ Deliverables for Faculty

Module	Deliverable
Introduction to Embedded Systems	Kit Code: OV-1515-KIT01-01-B
Programming Embedded Systems	Kit Name: Kit of 2 titles for OV-1515 Embedded Programming
Course	Classroom Presentations Trainer Guide Module PMs Master PM

5. Evaluation Strategy

The Evaluation Strategy for this course is as follows:

➤ Final Examination

5.1 Award of Grades

Overall Weighted Marks $\geq 40\%$ but $< 60\%$ qualifies for **PASS**

Overall Weighted Marks $\geq 60\%$ but $< 75\%$ qualifies for **CREDIT and**

Overall Weighted Marks $\geq 75\%$ qualifies for **DISTINCTION**

Note: To attain a PASS/CREDIT/DISTINCTION grade, a student should achieve at least 40% in Final Examination; otherwise he/she will be declared as 'Referred'.

5.2 Final Examination

Below is the list of modules to be covered in Final Examination along with their weightages:

Final Examination 1 (E1)

Marks: 50

Duration: 1 hour

Type: Objective

Sr. No.	Name of the Subject	Marks
1.	Introduction to Embedded Systems	25
2	Programming Embedded Systems	25

Final Examination 2 (R1)

Marks: 50

Duration: 2 hours

Type: Practical

Sr. No.	Name of the Subject	Marks
1.	Introduction to Embedded Systems	25
2	Programming Embedded Systems	25

Evaluation Details

OES – 50%

Practical – 50%

6. Classroom and Lab Pre-requisites

The requirements for conducting the sessions of the new curriculum have been described as follows:

6.1 Software Requirements

Microsoft Windows 10 or higher
Proview32 Simulator
µVision Simulator

6.2 Recommended Hardware

Intel Core i3/i5 Processor or higher
8 GB RAM or above
Color SVGA
500 GB Hard Disk space
Mouse
Keyboard, ARM board

6.3 Other Requirements (For Classroom)

Computer for giving demonstrations wherever applicable
LCD projector

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